Exchangeable Dyads: Model and Back-Transformation

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Assume `df_long_dim` contains one row per person-day and two rows per `coupleID:diaryday`. The exchangeability code uses `Idiff = +1/-1` to represent partner deviations from the dyad-level mean.

1 Data structure

Table 1: Example person-day structure for an exchangeable dyad

coupleID	userID	diaryday	couple_day	member	Idiff	is_partner1	is_partner2
31	31_1	0	31.0	1	1	1	0
31	31_1	1	31.1	1	1	1	0
31	31_1	2	31.2	1	1	1	0
31	31_2	0	31.0	2	-1	0	1
31	31_2	1	31.1	2	-1	0	1
31	31_2	2	31.2	2	-1	0	1

`Idiff` is constant within person and opposite within dyad. The dummy variables used in `glmmTMB`, `is_partner1` and `is_partner2`, must be constructed from the same `Idiff` coding.

2 glmmTMB model

```
model_exch_long_apim_glmmtmb <- glmmTMB(
  closeness ~ 1 +

    # Pooled fixed effects: no gender-specific intercepts or slopes.
    diaryday_c +
    provided_support_actor_cwp +
    provided_support_partner_cwp +
    provided_support_actor_cbp +
    provided_support_partner_cbp +

    # Dyad-level means and slopes.
    (1 + diaryday_c +
       provided_support_actor_cwp +
       provided_support_partner_cwp | coupleID) +

    # Partner deviations from the dyad-level effects.
    # This block is kept separate from the dyad-level block so that the
    # sum and difference components are independent.
```

```
    (0 + Idiff +
       I(Idiff * diaryday_c) +
       I(Idiff * provided_support_actor_cwp) +
       I(Idiff * provided_support_partner_cwp) | coupleID) +

    # Same-day residual interdependence:
    # homogeneous compound symmetry = one residual variance plus one
    # partner residual correlation.
    # Important: is_partner1 and is_partner2 must be constructed from the
    # same Idiff coding used above, e.g. is_partner1 = 1 when Idiff == 1
    # and is_partner2 = 1 when Idiff == -1.
    homcs(0 + is_partner1 + is_partner2 | coupleID:diaryday),

  # The residual variance is represented by the homcs block above.
  # Without dispformula = ~ 0, glmmTMB would add a second independent
  # residual variance on top of the dyadic residual covariance structure.
  # For families without a residual, dispformula should NOT be set to zero.
  dispformula = ~ 0,
  family = gaussian(),
  data = df_long_dim
)
```

The residual block could also be written as independent sum/difference terms:

```
(1 | coupleID:diaryday) +
(0 + Idiff | coupleID:diaryday)
```

With `Idiff = +1/-1`, this has the same parameter count as `homcs(...)`: one sum variance and one difference variance, which rotate back to one homogeneous residual variance and one residual covariance.

3 brms model

```
formula_exch_long_apim <- bf(
  closeness ~ 1 +

    # Pooled fixed effects.
    diaryday_c +
    provided_support_actor_cwp +
    provided_support_partner_cwp +
    provided_support_actor_cbp +
    provided_support_partner_cbp +

    # Dyad-level means and slopes.
    (1 + diaryday_c +
       provided_support_actor_cwp +
       provided_support_partner_cwp | coupleID) +

    # Partner deviations from the dyad-level effects.
    (0 + Idiff +
       I(Idiff * diaryday_c) +
       I(Idiff * provided_support_actor_cwp) +
       I(Idiff * provided_support_partner_cwp) | coupleID) +

    # Native residual correlation between the two partners on the same day.
    # We use unstr() instead of cs() because brms currently restricts cs()
    # residual correlations to be positive. With exactly two members per group
    # (dyads), unstr() is equivalent here because only one residual correlation
    # is estimated. For triads or larger groups, unstr() would no longer impose
    # the compound-symmetry constraint.
    unstr(time = member, gr = couple_day)

  # No sigma formula is specified here. Thus, sigma ~ 1 is implied,
  # meaning one homogeneous residual variance for both partners.
```

```r
  # If sigma is modelled with its own formula, brms estimates the sigma
  # coefficients on the log scale, so those coefficients must be exponentiated
  # before reporting them as residual SDs.
)

priors_exch_long_apim <- c(
  prior(normal(4, 2), class = "Intercept"),
  prior(normal(0, 2), class = "b"),
  prior(exponential(1), class = "sd"),
  prior(lkj(2), class = "cor"),
  prior(student_t(3, 0, 1.5), class = "sigma"),
  prior(lkj(2), class = "cortime")
)

model_exch_long_apim <- brm(
  formula = formula_exch_long_apim,
  data = df_long_dim,
  family = gaussian(link = identity),
  prior = priors_exch_long_apim,
  chains = 4,
  cores = 4,
  iter = 2000,
  warmup = 1000,
  seed = 123,
  file = file.path("brms_cache", "model_apim_ind_long_apim")
)
```

4 Back-transformation

The model estimates a sum block and an `Idiff` block. For exchangeable dyads, the full APIM random-effects matrix is recovered by rotating these two blocks:

$$\mathrm{Var}(A) = \mathrm{Var}(B) = \Sigma_{\mathrm{sum}} + \Sigma_{\mathrm{diff}}$$

$$\mathrm{Cov}(A, B) = \Sigma_{\mathrm{sum}} - \Sigma_{\mathrm{diff}}$$

This exact rotation assumes that the sum and difference random-effect blocks are independent.

The helper function `summarize_exchangeable_apim_brms()` is defined in `00_R_Functions/ReportModels.R`. It reconstructs the sum and `Idiff` covariance matrices for every posterior draw, rotates each draw into the full APIM partner-level matrix, and then summarizes the transformed covariance, SD, correlation, fixed-effect, and residual quantities.

For `glmmTMB`, the same back-transformation is algebraically possible, but interval estimation is less direct because there are no posterior draws. A `glmmTMB` workflow would usually transform parametric bootstrap draws or asymptotic draws from the fitted parameter covariance matrix.

```r
apim_results <- summarize_exchangeable_apim_brms(
  model_exch_long_apim,
  gr = "coupleID",
  idiff = "Idiff",
  term_labels = c("int", "day", "actor_wp", "partner_wp"),
  partner_labels = c("A", "B")
)
```

5 Variance-covariance matrix

```
kable(
  round(apim_results$full_covariance_matrix, 3),
  caption = "Back-transformed full APIM random-effects variance-covariance matrix",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 2: Back-transformed full APIM random-effects variance-covariance matrix

	A_int	A_day	A_actor_wp	A_partner_wp	B_int	B_day	B_actor_wp	B_partner_wp
A_int	1.176	0.011	0.095	0.018	-0.275	-0.002	-0.064	0.004
A_day	0.011	0.000	0.001	0.000	-0.002	0.000	-0.001	0.000
A_actor_wp	0.095	0.001	0.024	0.004	-0.064	-0.001	-0.018	-0.002
A_partner_wp	0.018	0.000	0.004	0.012	0.004	0.000	-0.002	0.003
B_int	-0.275	-0.002	-0.064	0.004	1.176	0.011	0.095	0.018
B_day	-0.002	0.000	-0.001	0.000	0.011	0.000	0.001	0.000
B_actor_wp	-0.064	-0.001	-0.018	-0.002	0.095	0.001	0.024	0.004
B_partner_wp	0.004	0.000	-0.002	0.003	0.018	0.000	0.004	0.012

6 SD table

```
kable(
  apim_results$sd_summary,
  digits = 3,
  caption = "Back-transformed APIM random-effect standard deviations with 95% credible intervals"
)
```

Table 3: Back-transformed APIM random-effect standard deviations with 95% credible intervals

parameter	estimate	est_error	q_low	q_high
A_actor_wp	0.153	0.018	0.119	0.190
A_day	0.018	0.001	0.016	0.020
A_int	1.083	0.057	0.976	1.196
A_partner_wp	0.107	0.021	0.065	0.147
B_actor_wp	0.153	0.018	0.119	0.190
B_day	0.018	0.001	0.016	0.020
B_int	1.083	0.057	0.976	1.196
B_partner_wp	0.107	0.021	0.065	0.147

7 Correlation table

```
kable(
  round(apim_results$full_correlation_matrix, 3),
  caption = "Back-transformed APIM random-effect correlation matrix",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 4: Back-transformed APIM random-effect correlation matrix

	A_int	A_day	A_actor_wp	A_partner_wp	B_int	B_day	B_actor_wp	B_partner_wp
A_int	1.000	0.579	0.569	0.153	-0.233	-0.128	-0.380	0.038
A_day	0.579	1.000	0.362	0.144	-0.128	-0.134	-0.207	0.045
A_actor_wp	0.569	0.362	1.000	0.253	-0.380	-0.207	-0.757	-0.105
A_partner_wp	0.153	0.144	0.253	1.000	0.038	0.045	-0.105	0.241
B_int	-0.233	-0.128	-0.380	0.038	1.000	0.579	0.569	0.153
B_day	-0.128	-0.134	-0.207	0.045	0.579	1.000	0.362	0.144
B_actor_wp	-0.380	-0.207	-0.757	-0.105	0.569	0.362	1.000	0.253
B_partner_wp	0.038	0.045	-0.105	0.241	0.153	0.144	0.253	1.000

8 Final reporting summary

Table 5: Compact reporting summary

section	parameter	estimate	est_error	q_low	q_high
Fixed effects	Intercept	5.111	0.064	4.990	5.235
Fixed effects	diaryday_c	0.004	0.001	0.001	0.006
Fixed effects	provided_support_actor_cwp	0.300	0.013	0.274	0.327
Fixed effects	provided_support_partner_cwp	0.149	0.015	0.120	0.179
Fixed effects	provided_support_actor_cbp	1.304	0.088	1.136	1.477
Fixed effects	provided_support_partner_cbp	0.402	0.082	0.232	0.560
Back-transformed random-effect SDs	A_actor_wp	0.153	0.018	0.119	0.190
Back-transformed random-effect SDs	A_day	0.018	0.001	0.016	0.020
Back-transformed random-effect SDs	A_int	1.083	0.057	0.976	1.196
Back-transformed random-effect SDs	A_partner_wp	0.107	0.021	0.065	0.147
Back-transformed random-effect SDs	B_actor_wp	0.153	0.018	0.119	0.190
Back-transformed random-effect SDs	B_day	0.018	0.001	0.016	0.020
Back-transformed random-effect SDs	B_int	1.083	0.057	0.976	1.196
Back-transformed random-effect SDs	B_partner_wp	0.107	0.021	0.065	0.147
Back-transformed random-effect correlations	A_actor_wp with A_day	0.364	0.102	0.150	0.550
Back-transformed random-effect correlations	A_actor_wp with A_int	0.572	0.085	0.386	0.722
Back-transformed random-effect correlations	A_day with A_int	0.579	0.054	0.467	0.678
Back-transformed random-effect correlations	A_partner_wp with A_actor_wp	0.265	0.157	-0.026	0.574
Back-transformed random-effect correlations	A_partner_wp with A_day	0.147	0.130	-0.112	0.399
Back-transformed random-effect correlations	A_partner_wp with A_int	0.157	0.119	-0.077	0.387
Back-transformed random-effect correlations	B_actor_wp with A_actor_wp	-0.759	0.153	-0.991	-0.414
Back-transformed random-effect correlations	B_actor_wp with A_day	-0.209	0.112	-0.418	0.024
Back-transformed random-effect correlations	B_actor_wp with A_int	-0.382	0.102	-0.573	-0.182
Back-transformed random-effect correlations	B_actor_wp with A_partner_wp	-0.110	0.159	-0.414	0.199
Back-transformed random-effect correlations	B_actor_wp with B_day	0.364	0.102	0.150	0.550
Back-transformed random-effect correlations	B_actor_wp with B_int	0.572	0.085	0.386	0.722
Back-transformed random-effect correlations	B_day with A_actor_wp	-0.209	0.112	-0.418	0.024
Back-transformed random-effect correlations	B_day with A_day	-0.133	0.120	-0.364	0.105
Back-transformed random-effect correlations	B_day with A_int	-0.128	0.085	-0.291	0.039
Back-transformed random-effect correlations	B_day with A_partner_wp	0.046	0.135	-0.214	0.305
Back-transformed random-effect correlations	B_day with B_int	0.579	0.054	0.467	0.678
Back-transformed random-effect correlations	B_int with A_actor_wp	-0.382	0.102	-0.573	-0.182
Back-transformed random-effect correlations	B_int with A_day	-0.128	0.085	-0.291	0.039
Back-transformed random-effect correlations	B_int with A_int	-0.232	0.091	-0.405	-0.054
Back-transformed random-effect correlations	B_int with A_partner_wp	0.039	0.126	-0.207	0.279
Back-transformed random-effect correlations	B_partner_wp with A_actor_wp	-0.110	0.159	-0.414	0.199
Back-transformed random-effect correlations	B_partner_wp with A_day	0.046	0.135	-0.214	0.305
Back-transformed random-effect correlations	B_partner_wp with A_int	0.039	0.126	-0.207	0.279
Back-transformed random-effect correlations	B_partner_wp with A_partner_wp	0.263	0.381	-0.522	0.972
Back-transformed random-effect correlations	B_partner_wp with B_actor_wp	0.265	0.157	-0.026	0.574
Back-transformed random-effect correlations	B_partner_wp with B_day	0.147	0.130	-0.112	0.399
Back-transformed random-effect correlations	B_partner_wp with B_int	0.157	0.119	-0.077	0.387
Residual structure	residual_variance	0.878	0.012	0.855	0.902
Residual structure	same_day_residual_correlation	0.038	0.014	0.008	0.064
Residual structure	same_day_residual_covariance	0.033	0.013	0.007	0.056
Residual structure	sigma	0.937	0.007	0.925	0.950